

TOSAN TABLET 50MG

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Each tablet contains Losartan Potassium 50.0mg

DESCRIPTION

White oval biconvex film coated tablet with half score and letter C, L on one side and figure 50 on the other side.

Pharmacodynamics

Angiotensin II is a potent vasoconstrictor, the primary vasoactive hormone of the renin-angiotensin system and an important component in the pathophysiology of hypertension. Angiotensin II also stimulates aldosterone secretion by the adrenal cortex and smooth muscle cell proliferation. Losartan and its active carboxylic acid metabolite are specific angiotensin II receptor (type AT₁) antagonist. AT₁ receptor can be found in many tissues in the body (e.g. vascular smooth muscle, adrenal gland, kidneys and heart). Losartan inhibits systolic and diastolic pressor responses to angiotensin II infusions by binding selectively to the AT₁ receptor, resulting in blockage of vasoconstriction and the release of aldosterone.

The removal of angiotensin II negative feedback on renin secretion during treatment with losartan gave rise to plasma renin activity and this leads to increase angiotensin II in plasma. Even with these increases, the antihypertensive activity and suppression of plasma aldosterone concentration are maintained, indicating an effective angiotensin-II receptor blockade.

Losartan does not block or bind to other hormone receptors or ion channels which are important in cardiovascular regulation. Generally, losartan has been shown to block responses to angiotensin I and angiotensin II without affecting responses to bradykinin. It also does not inhibit Angiotensin-converting enzyme (ACE or kininase II), the enzyme that degrades bradykinin.

In non-diabetic hypertensive patients with proteinuria, the administration of losartan significantly reduces proteinuria, fractional excretion of albumin and IgG, besides maintains glomerular filtration rate and reduces filtration fraction in these patients. Losartan also caused a transient increase in urinary uric acid excretion (usually < 24 micromol) which was persistent in chronic therapy.

When losartan is administered in doses up to 150mg once daily, it does not cause clinically important changes in fasting triglycerides, total cholesterol or HDL-cholesterol in patients with hypertension. The same dose of losartan also has no effect on fasting glucose levels.

Losartan does not have any effect on the autonomic reflexes and sustained effect on the plasma norepinephrine. The antihypertensive effect of losartan is dose dependent.

Pharmacokinetics

Absorption

Following oral administration, losartan is well absorbed and undergoes substantial first-pass metabolism by cytochrome P450 enzymes in the liver, thus forming an active carboxylic acid metabolite and other inactive metabolites. The systemic

bioavailability of losartan is about 33%. Peak plasma concentrations of losartan and its active metabolite occur in about 1 hour and 3 to 4 hours respectively after an oral dose. There was no significant clinical effect on losartan plasma concentration when the drug is administered with a standardized meal.

Distribution

Both losartan and its active metabolite are more than 98% bound to plasma proteins, primarily albumin. The volume of distribution of losartan and the active metabolite is about 34 litres and 12 litres respectively. Losartan crosses the blood-brain barrier poorly.

Biotransformation

About 14% of an orally-administered dose of losartan is converted to active metabolite. In addition to the active metabolite, inactive metabolites are also formed, including two major metabolites formed by hydroxylation of the butyl side chain and a minor metabolite, an N-2 tetrazole glucuronide.

Elimination

Plasma clearance of losartan and its active metabolite is about 600mL/min and 50mL/min while renal clearance is about 74mL/min and 26mL/min respectively. About 4% of the dose is excreted as unchanged in the urine and about 6% is excreted as active metabolite. Both biliary and urinary excretions contribute to the elimination of losartan and its metabolites.

Following oral administration, plasma concentrations of losartan and its active metabolite decline polyexponentially with a terminal half-life of about 2 hours and 6-9 hours respectively. Losartan and its active metabolite do not accumulate significantly in plasma upon repeated once-daily dosing.

Neither losartan nor its active metabolite can be removed by haemodialysis.

INDICATIONS

Tosan Tablet is indicated for the treatment of:

- 1) Hypertension.
- 2) Hypertension with Left Ventricular Hypertrophy - reduce the risk of stroke in patient with hypertension and left ventricular hypertrophy.
- 3) Nephropathy in Type 2 Diabetic Patients (Macroalbuminuria) with an elevated serum creatinine and proteinuria (urinary albumin to creatinine ratio more than 300mg/g) - delay the progression of diabetic nephropathy as measured by a reduction in the combined incidence of doubling the serum creatinine, end stage renal disease (need for dialysis or renal transplantation) or death; and to reduce proteinuria.

RECOMMENDED DOSAGE

Tosan Tablet may be administered with or without food and may be taken concomitantly with other antihypertensive agents.

Hypertension

The starting and maintenance dose is 50mg (1 tablet) once daily and the maximal antihypertensive effect is reached 3-6 weeks

after initiation of therapy. Some patients may require a higher dose of up to 100mg (2 tablets of **Tosan Tablet**) once daily.

For patients with intravascular volume depletion (e.g. patients treated with high dose of diuretics) a starting dose of 25mg once daily is advisable.

For patients with a history of hepatic impairment, a lower dose is recommended.

No initial dosage adjustment is necessary for elderly patients or for patients with renal impairment, including patients on dialysis.

Losartan is not recommended to use in children with hepatic impairment.

Hypertensive Patients with Left Ventricular Hypertrophy

The starting dose is 50mg (1 tablet) once daily. A low dose of hydrochlorothiazide may be added and/or the dose of **Tosan Tablet** may be increased to 100mg (2 tablets) once daily based on the blood pressure response.

Renal Protection in Type 2 Diabetic Patients with Proteinuria and Hypertension

The starting dose is 50mg (1 tablet) once daily. The dose may be increased to 100mg (2 tablets) once daily depending on the blood pressure response. **Tosan Tablet** may be administered concurrently with other antihypertensive agents (e.g. diuretics, calcium channel blockers, alpha- or beta-blockers and centrally acting agents) as well as with insulin or with other commonly used hypoglycaemic agents (e.g. sulfonylureas, glitazones and glucosidase inhibitors).

Mode of Administration

For oral administration.

CONTRAINDICATIONS

Tosan Tablet is contraindicated in patients who are hypersensitive to any component of this product and in pregnant women.

WARNINGS AND PRECAUTIONS

Hypersensitivity

Angioedema.

Hypotension and Electrolyte / Fluid Imbalance

In patients who are intravascularly volume-depleted (e.g. those treated with high-dose of diuretics), symptomatic hypotension may occur. This condition should be corrected prior administration of **Tosan Tablet** or a lower starting dose is recommended.

Electrolyte imbalance is common in elderly, patients with renal impairment, diabetic or non-diabetic patients and should be treated accordingly. High incidence of hyperkalaemia has been reported in such patients taking losartan. Therefore, serum-potassium level should be monitored in this group of patients and it is not advisable to use potassium supplements or potassium-sparing diuretics concurrently with losartan.

Hepatic Function Impairment

A lower dose should be considered for patients with history of hepatic impairment as pharmacokinetics literature shown that,

plasma concentration of losartan increase significantly in cirrhotic patients.

Renal Function Impairment

As a consequences of inhibiting the renin-angiotensin-aldosterone system, changes in renal function including renal failure have been reported particular, in patients whose renal function is dependent on the renin-angiotensin-aldosterone system such as those patients with severe cardiac insufficiency or pre-existing renal dysfunction.

Caution is required in patients with significant renal disease and renal transplant recipients as there have been reports of anaemia developing in such patients.

Other drugs including **Tosan Tablet** that affect the renin-angiotensin-aldosterone system may increase blood urea and serum creatinine in patients with bilateral renal artery stenosis or stenosis of the artery to a solitary kidney.

These changes in renal function may be reversible upon discontinuation of therapy.

Use in Elderly

There was no age-related difference in the efficacy or safety profile of losartan.

Paediatric use

In paediatric patients who are intravascularly volume depleted, these conditions should be corrected prior administration of **Tosan Tablet**.

Tosan Tablet is not recommended in paediatric patients with glomerular filtration rate below 30mL/min/1.73m², in paediatric patients with hepatic impairment or in neonates as no data is available.

Interaction with other medicaments

Concomitant use of drugs which retain potassium or may increase potassium levels such as potassium-sparing diuretics (e.g. spironolactone, triamterene, and amiloride), potassium supplements or salt substitutes containing potassium may result in hyperkalaemia. Co-administration is not advisable but if concomitant use of such product is imperative, special caution and regular monitoring of the plasma potassium are required.

Concurrent use with sympathomimetic drugs may reduce the antihypertensive effect of **Tosan Tablet**.

No drug interactions of clinical significance have been identified with hydrochlorothiazide, digoxin, warfarin, cimetidine, phenobarbital, ketoconazole and erythromycin. Rifampin and fluconazole have been reported to reduce the level of losartan's active metabolite.

Lithium

As with other drugs which affect the excretion of sodium, lithium excretion may be reduced. Therefore, serum lithium levels should be monitored carefully if lithium salts are to be co-administered with angiotensin II receptor antagonist.

Non-steroidal anti-inflammatory drugs (NSAIDs)

When angiotensin II receptor antagonist is administered simultaneously with NSAIDs (i.e. selective cyclooxygenase-2

inhibitors (COX-2 inhibitors), acetylsalicylic acid more than 3g/day and non-selective NSAIDs), attenuation of the antihypertensive effects may occur.

In some patients, especially in the elderly and patients with compromised renal function, caution should be exercised when angiotensin II receptor antagonists are co-administered with NSAIDs, including selective COX-2 inhibitors. This is because co-administration may increase risk of worsening renal function including possible acute renal failure and increased in serum potassium level, especially in patients with poor pre-existing renal function. However, these effects are usually reversible.

Pregnancy and lactation

Pregnancy

In humans, foetal renal perfusion which depends on the development of the renin-angiotensin-aldosterone system begins in the second trimester of pregnancy. Therefore, the risk to the foetus increases if **Tosan Tablet** is administered during the second or third trimesters of pregnancy.

Losartan act directly on the foetal's renin-angiotensin system and causes injury including hypotension, oliguria, hyperkalaemia and even death in the developing foetus. When pregnancy is detected, treatment with **Tosan Tablet** should be discontinued immediately.

Lactation

It is not known whether losartan is excreted into human's milk. Due to the potential of adverse effects on the nursing infant, a decision should be made whether to discontinue nursing or discontinue the drug, taking into account the importance of the drug to the mother.

ADVERSE REACTIONS

Tosan Tablet is generally well tolerated by patients and the side effect is usually mild and transient in nature.

The adverse reaction for paediatric group appears to be similar to that seen in adult patients.

Hypersensitivity: Anaphylactic reactions, angioedema including swelling of the larynx and glottis causing airway obstruction and/or swelling of the face, lips, pharynx and/or tongue, vasculitis or angitis including Henoch-Schoenlein purpura with abdominal pains, gastrointestinal haemorrhage and nephritis has been reported rarely. Some patients may previously experienced angioedema with other drugs including ACE inhibitors.

Gastrointestinal: Hepatitis (reported rarely), liver function abnormalities, vomiting, diarrhoea, dyspepsia, severe acute hepatotoxicity, dry mouth, flatulence, colitis ulcerative, cholelithiasis and acute pancreatitis. Slight increase in liver enzymes may occur and usually receded when treatment is discontinued.

Haematologic: Anaemia, thrombocytopenia (reported rarely), pancytopenia, cholestasis and neutropenia.

Musculoskeletal: Myalgia, arthralgia, convulsion, back pain and pain in extremities, arthritis, rare cases of rhabdomyolysis have been reported in patients receiving angiotensin II receptor antagonist.

Nervous System / Psychiatric: Migraine, dysgeusia (taste disturbance), sleep disturbances, apathy, hallucinations, memory impairment, disorientation, depression, anxiety, fatigue, insomnia, psychiatric disorder, dizziness, nausea, headache, abdominal pain, epistaxis, asthenia and malaise.

Respiratory: Cough, nasal congestion, pharyngitis, upper respiratory infection, sinus disorder (rarely).

Skin: Urticaria, pruritus, erythroderma, rash, psoriasis, alopecia, photosensitivity, cyanosis.

Metabolic and nutrition: Hyperkalaemia, hypoglycaemia, dehydration, hyponatraemia, gout.

Cardiologic: pulmonary hypertension, palpitations, tachycardia, hypotension, chest pain, atrial and ventricular fibrillation and coronary artery disease.

Nephrology: Renal failure and acute renal failure.

SYMPTOMS & TREATMENT OF OVERDOSAGE

Limited data are available in regard to overdosage in humans. The most likely manifestation of overdosage would be hypotension and tachycardia; bradycardia may occur due to parasympathetic (vagal) stimulation.

If severe hypotension occurred, supportive treatment should be instituted. Patients should be placed in supine position and rapid infusion of the necessary quantity of physiological saline solution may be given and the vital sign be monitored. **Tosan Tablet** therapy should be withdrawn and the patient be placed under close medical supervision.

Neither losartan nor the active metabolite can be removed by haemodialysis.

STORAGE CONDITIONS

Store in a cool dry place, below 30 °C. Protect from light.

Keep out of reach of children. Jauhi daripada kanak-kanak.

SHELF LIFE

Please refer to the outer box label.

PACKING/PACK SIZES

Blister pack of 10 tablets per strip. Pack in a box of 3 x 10, 5 x 10, 10 x 10, 50 x 10 tablets.

MANUFACTURED BY

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